Health economic value and choice in the age of artificial intelligence

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Supporting information

Supplementary Tables 1–3, Extended Data Figs. 1–4 and Supplementary References.

Supplementary Table 1 **| Minimum comparison record.** Proposed fields, not a validated scale. Health capability, ICF, WHO-5, MCDA and decision-quality research inform the distinctions[1–4] (https://www.who.int/standards/classifications/international-classification-of-functioning-disability-and-health).

Domain	Record	Interpretation
Decision	Feasible options; ability to refuse; common horizon	Distinguish nominal options from real opportunities
Resources and time	Net income change; total work; displaced activities	Record where saved time is used
Health and functioning	Relevant symptoms, function or clinical outcomes	Interpret against baseline and clinical context
Experience	Enjoyment, relief or distress; during and after	Separate immediate responses from sustained change
Capabilities and agency	Independent performance; valued activities; discretion	Include accessibility gains from assistance
Evidence and appraisal	Population; comparator; horizon; uncertainty; goals	Retain unknowns and differences in priorities

ICF, International Classification of Functioning, Disability and Health; WHO-5, World Health Organization-Five Well-Being Index; MCDA, multicriteria decision analysis.

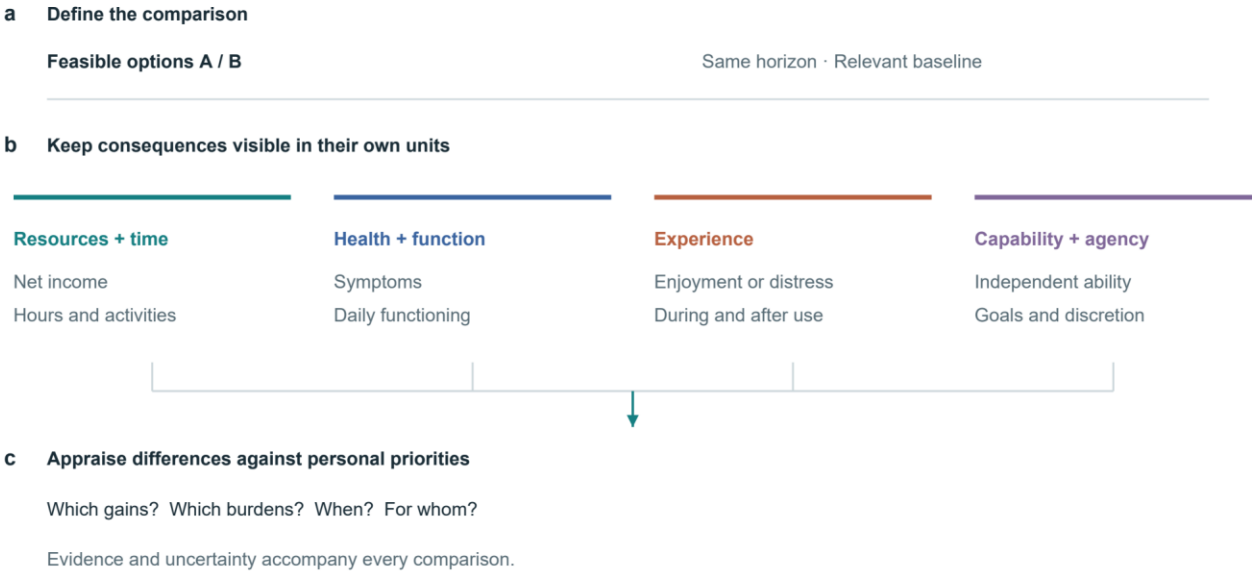
Supplementary Table 2 **| Different levels of AI outcomes.** Studies illustrate task efficiency, experience and judgement; their designs and endpoints differ.

Setting and design	Main finding	Interpretive scope
Professional writing[5] Randomized; n = 453	Faster task completion and better output quality	Full-day allocation and health not measured
Customer support[6] Staggered rollout; n = 5,172	About 15% higher resolutions per hour; outcome availability varies	Causal assumptions; allocation of saved time not measured
AI companionship[7] Seven-day Study 3	Immediate loneliness relief; persistent pre-interaction benefit not demonstrated	Differential attrition; control not time-matched
Medical scenarios[8] n = 1,298	Control versus pooled AI: identification OR 1.76 (95% CI 1.45–2.13)	Scenario performance, not clinical morbidity

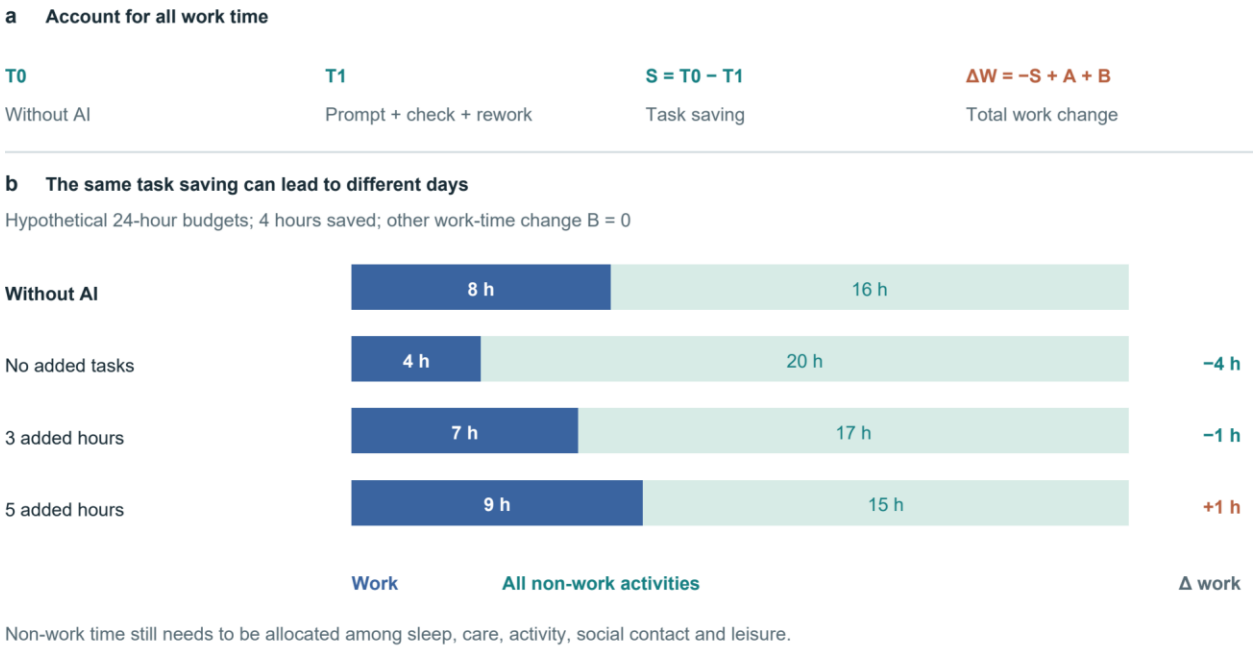
AI, artificial intelligence; OR, odds ratio; CI, confidence interval.

Supplementary Table 3 **| Testable propositions and disconfirming observations.** Each comparison requires prespecified estimands and outcomes.

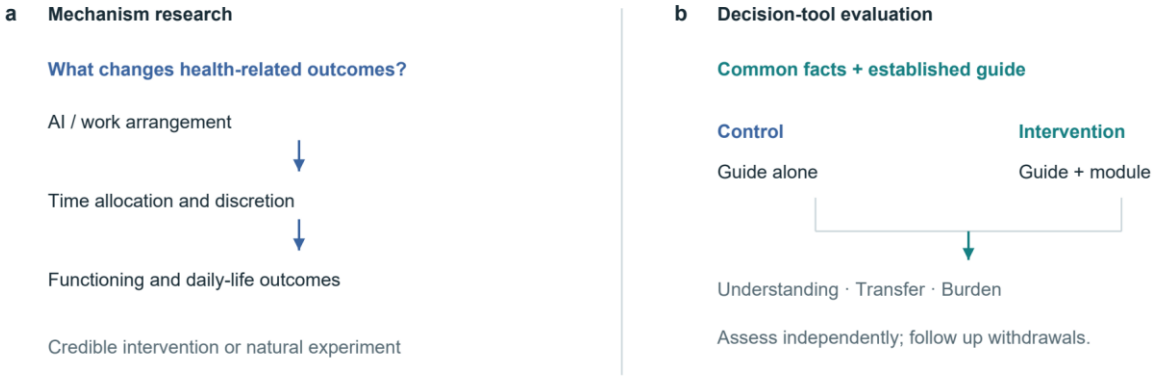
Proposition	Key comparison	Potentially disconfirming result
Discretion affects allocation of AI time savings	Similar task gains; credible differences in control over workload	No meaningful allocation difference, with sufficiently precise estimates
Immediate relief and sustained benefit may diverge	Matched comparator; repeated pre-use, post-use and follow-up measures	Stable pre-use improvement comparable to immediate relief
The module improves decision quality	Same guide, with or without the added module	No gain in understanding or transfer, or burden exceeds benefit



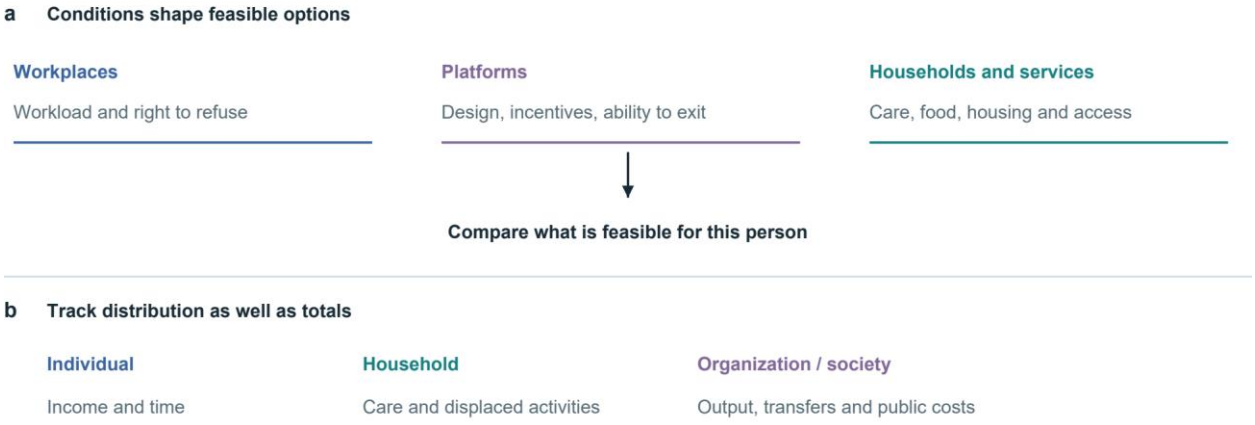
Extended Data Fig. 1 | **A framework for comparing health-related consequences.** Feasible alternatives and a common horizon define the comparison. Resources and time, health and functioning, experience, and capabilities and agency remain distinct. Priorities inform appraisal. Connections represent analytical steps, not estimated causal effects.



Extended Data Fig. 2 **| Task savings and the allocation of a day. a,** The accounting identity includes prompting, verification and rework in T1, additional tasks in A and other work-time changes in B. **b,** Hypothetical 24-hour budgets with an 8-hour baseline workday, a 4-hour task saving and B = 0. Non-work time is not assigned a health effect.



Extended Data Fig. 3 **| Two research questions and their evaluation paths.** Mechanism research explains how outcomes arise; tool evaluation tests the added comparison module. An active comparator gives both groups the same facts and established guide. The proposed design is not a completed trial.



Extended Data Fig. 4 | **Conditions of choice and distribution of consequences.** Workplace, platform and household conditions shape feasible options. Income, time and care burdens may accrue to different people. Arrows organize evaluation questions rather than depict estimated causal effects.

Supplementary references

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